

Robofreaks CRO Team



Hardware

Chassis

- Fully custom chassis designed in Fusion 360, 3D printed in PLA
- Mounts for our own PCBs, sensors, motors and their drivers
- Designed to be quickly repairable and lightweight

Motors

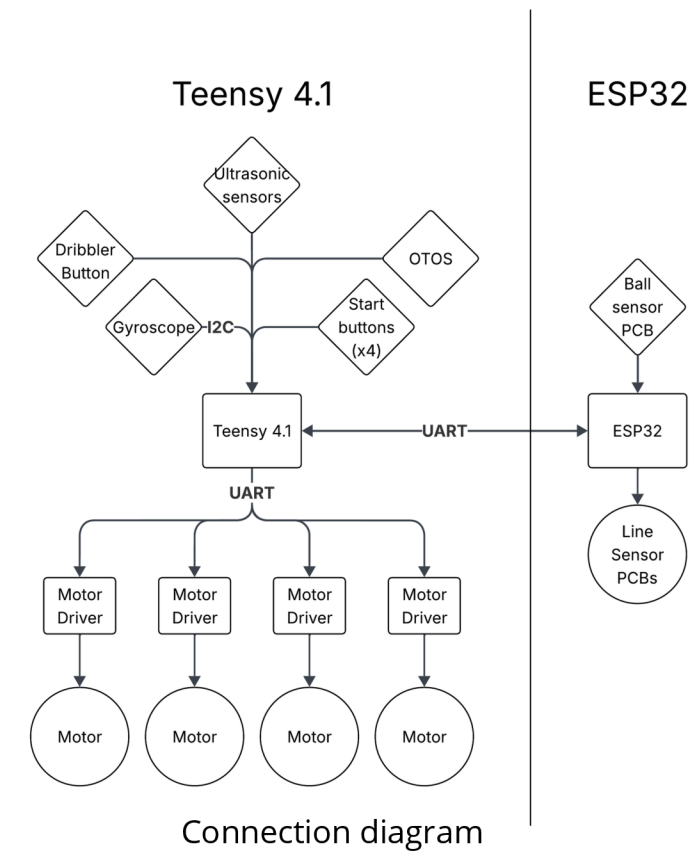
- 4x BLDC motors built in to the wheels of the robots
- Controlled by 4 Makerbase ODrive Mini FOC motor drivers

Power

- 1x 4S LiPo battery made by us from 18650 cells in the bottom part of the chassis
- Is lowered to 5 V and 3.3 V for powering sensors, micro controllers and other systems on the robot.

Controll

- Teensy 4.1 - main controller, handles the robot's movement and talks to other controllers
- ESP32 - extracting line and ball sensor data and sending over to the Teensy
- Makerbase ODrive motor controllers - controlled over UART from the Teensy, allow a high degree of precision in motor controll

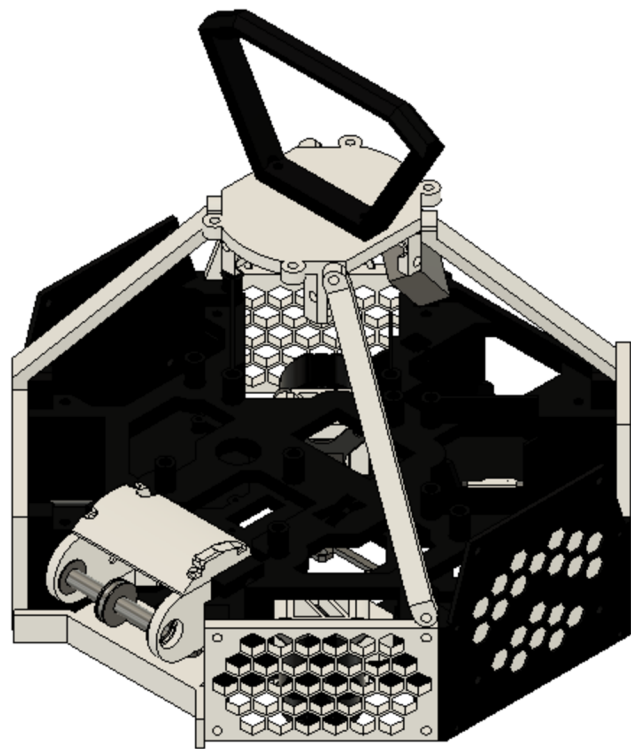


Sensors

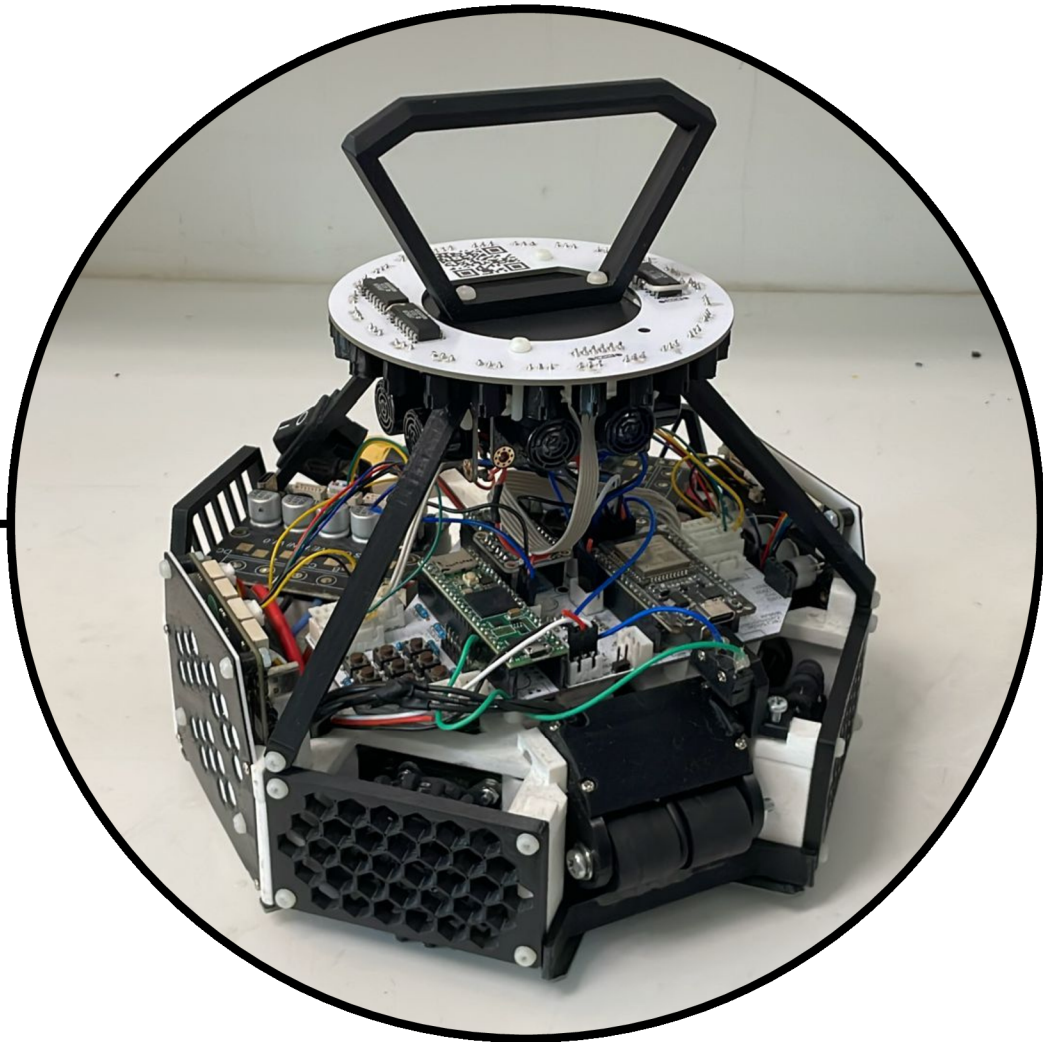
- 1x ball sensor PCB designed by us consisting of 24 TSSP4038 sensors connected to multiplexers
- 4x line sensor PCB each consisting of 16 sensors and 16 RGB LEDs connected to multiplexers
- 4x DFRobot analog ultrasonic distance sensors used for wall distances
- 1x SparkFun OTOS (Optical Tracking Odometry Sensor) used for precise positioning on the field
- 1x button above the dribbler used for knowing if the ball is in the dribbler
- 1x Adafruit BNO085 IMU used for robot direction readings

Dribbler

- The dribbler consists of a simple BLDC motor and its ESC in the chassis connected to the ball roller suspended in two bearings and connected to the motor via an O-ring serving as a belt drive
- Not yet perfect, but can keep the ball in and send it to the other side of the pitch
- The rubber used are rubber vibration isolating washers
- The button above the dribbler activates when the dribbler is lifted by a ball



Chassis design in Fusion



Team



Marin Rođak

- Team captain
- Programmer, planned and wrote all the code on the robot
- Experience:
 - RoboCupJunior European Bari (Rescue line) 2025. - 4th place
 - RoboCupJunior Salvador (Rescue line) 2025. - 4th place place
 - RoboCupJunior Americas (Rescue line) 2025. - 1st place

Patricija Radiković

- Team co-captain
- Hardware and electronics designer, designed the robot's chassis and PCB
- Experience:
 - RoboCupJunior European Hannover (CoSpace rescue) 2024. - 3rd place
 - Robocup Asia-Pacific Qingdao (coSpace) - 3rd place super teams
 - RoboCupJunior Americas (Rescue line) 2025. - 1st place



Ivan Kolarić

- Team mentor



Software

Language

- All code was written in Arduino

Structure

- The program has a very simple structure which consists of checking all sensors, updating their data and going in the best direction to intercept the ball

Ball detection

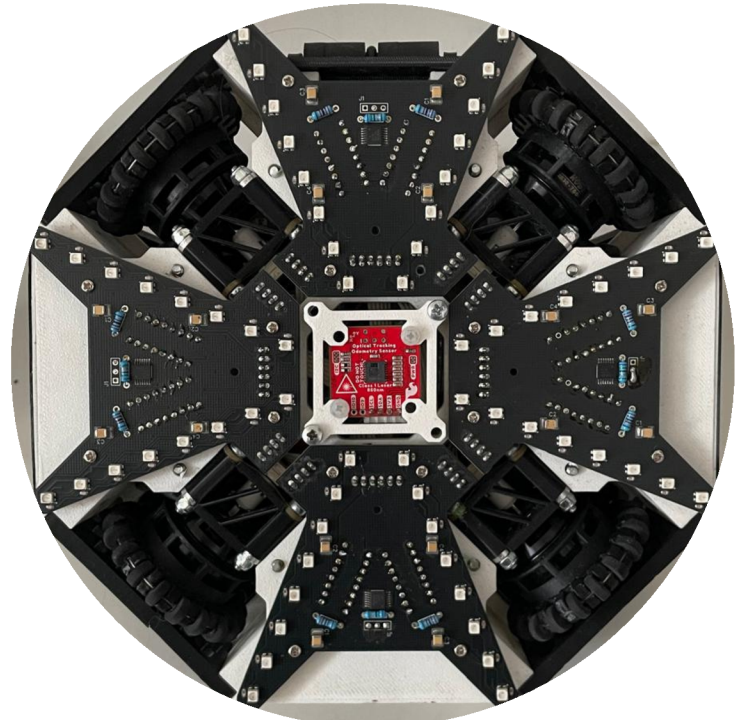
- ESP32 reads 24 TSSP4038 IR sensors through multiplexers
- Longest contiguous arc of active sensors is found (handles 0°/360° wrap-around)
- Circular mean of the arc gives a stable ball angle in degrees

Line detection

- 64 sensors normalized per-sensor using two-stage EEPROM calibration (carpet vs. white line)
- Grouped into 8 directional closeness representing 8 lines of sensors - values and sent to Teensy over UART
- Teensy runs quickAvoidLine() — records entry direction and drives back untill no line is visible

Movement

- All movement is field-relative: the desired forward/right velocity vector is rotated by the current IMU heading before being distributed to the four motors
- A PD rotation controller runs simultaneously on every loop iteration, adding a heading correction to each motor to maintain the target angle
- For navigation to a field coordinate, robotToPos() uses a PD controller where the proportional term is Euclidean distance and the derivative term is the robot's OTOS velocity projected onto the direction of travel
- Motor desired velocities and accelerations and sends them to the motor drivers via UART



Bottom of the robot (line sensors and OTOS)

Tactics

- One robot is always playing an attack role and the other is playing as a defender
- The defending robot is only trying to stop the ball from going into the goal and the attacker is trying to get behind the ball as fast as possible and push it into the opponents goal
- The defending robot will try to kick the ball over the center if it is alone on the field

Future

- We haven't had time to implement some features that are available in software, such as using the dribbler to shoot the ball, communication between robots and implementing a kicker
- There are lots of bugs and issues with the current code which need to be fixed to get the maximum possible performance out of all systems